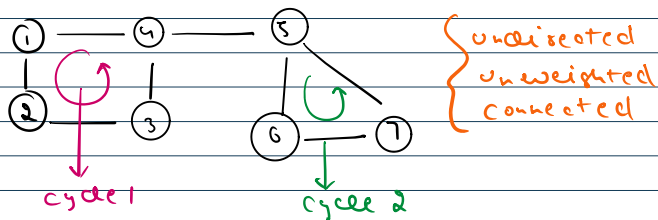


Graphs



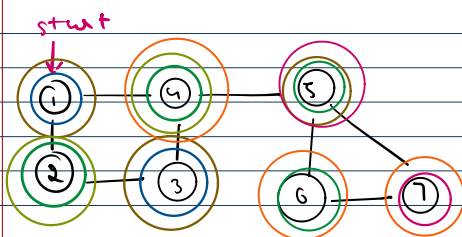
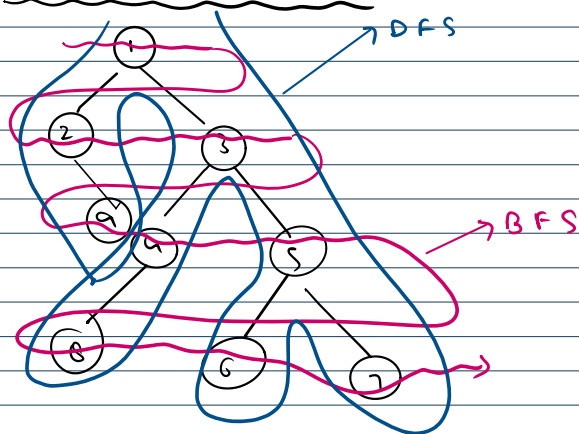
Traverse $\Rightarrow$ Go to all vertices and edges

Traverse $\begin{cases} \rightarrow \text{BFS (queue)} \\ \rightarrow \text{DFS (stack)} \end{cases}$

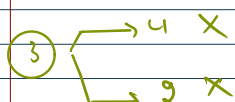
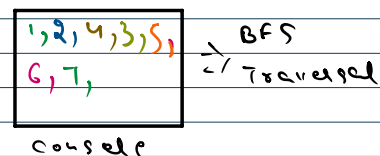
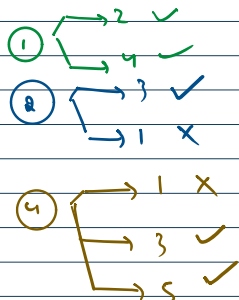
5 steps for a graph

1. Remove a vertex from queue/stack
2. Ignore if visited
3. mark as visited
4. Self work
5. Add unvisited nodes to the queue/stack

Breadth First Search

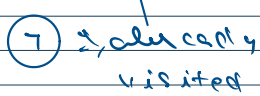
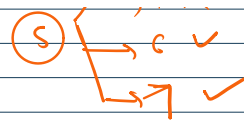


1. Remove
2. Ignore
3. mark as visited
4. Self work
5. Add unvisited nodes



3 $\Rightarrow$ already visited $\rightarrow$ there is a cycle





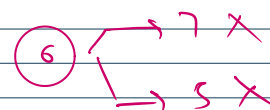
⇒ visited
↓
HashSet

console

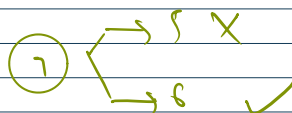
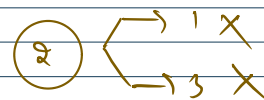
$$\xrightarrow{2} Ags$$

11. Remove
- ~~12. Ignore~~
31. mark as visited
41. Self work
51. Add unvisited
nbrs

Stack
↓
LIFO
↓
Last In
First out



⑥ $\Rightarrow$ already $\Rightarrow$ cell visited



2 $\Rightarrow$ already visited $\Rightarrow$ cycle